AUTO LIGHT SENSOR

id091800010600

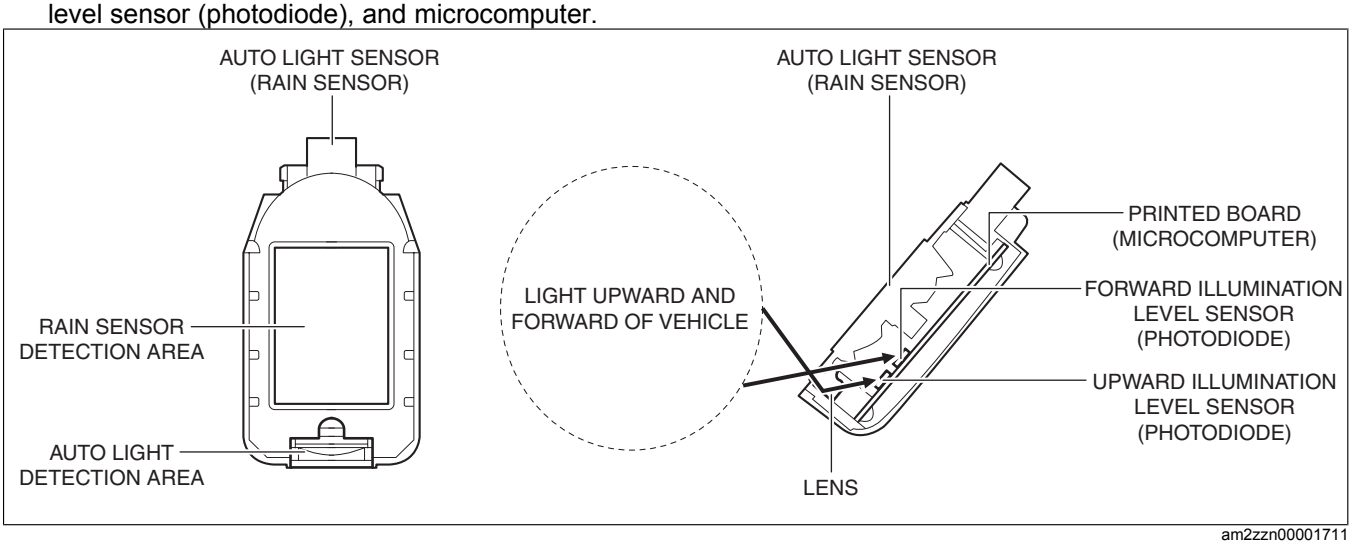
Purpose
- The auto light sensor detects the illumination level above and in front of the vehicle and outputs it to the BCM.

Function
- The auto light sensor uses the upward illumination level sensor to detect the illumination level above the vehicle, and uses the forward illumination level sensor to detect the illumination level in front of the vehicle.

Illumination level adjustment function
- The illumination level sensitivity can be switched between 5 levels using the Mazda Modular Diagnostic System (M-MDS) or center display (with center display). (See LIGHTING SYSTEM PERSONALIZATION.)

On-board diagnostic function
- If there is a malfunction in the auto light sensor, notifies the BCM and outputs a DTC from the BCM.
- For details on DTCs, refer to the DTC table [ON-BOARD DIAGNOSTIC SYSTEM [BODY CONTROL MODULE (BCM)]]. (See ON-BOARD DIAGNOSTIC SYSTEM [BODY CONTROL MODULE (BCM)].)

Construction
- The auto light sensor is integrated with the rain sensor as a single unit.
- The auto light sensor consists of the lens, upward illumination level sensor (photodiode), forward illumination level sensor (photodiode), and microcomputer.

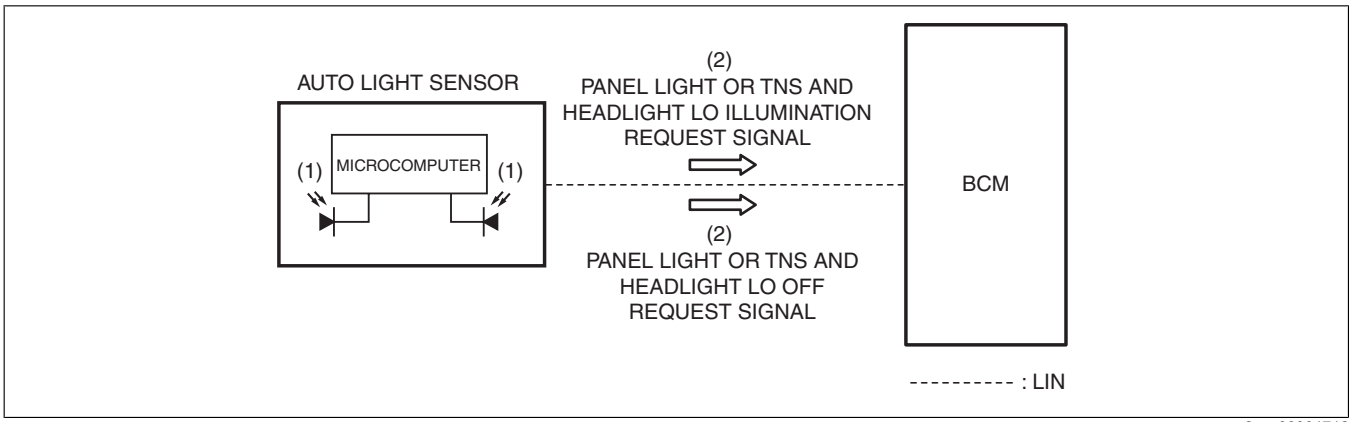


am2zzn00001711

Operation

Caution
- **The auto light sensor cannot detect the illumination level correctly under the following conditions, which could cause the auto light system to operate improperly.**
  - **— Stickers or labels are attached to the auto light sensor detection area on the windshield.**
  - **— The auto light sensor detection area on the windshield is dirty.**

1. When the ignition is switched ON (engine off or on), the auto light sensor detects (1) the illumination level above and in front of the vehicle.
2. The auto light sensor outputs (2) a panel light or TNS and headlight LO illumination request signal and panel light or TNS and headlight LO off request signal to the BCM via LIN communication according to the detected illumination level.



am2zzn00001712

Fail-safe

- Not applicable